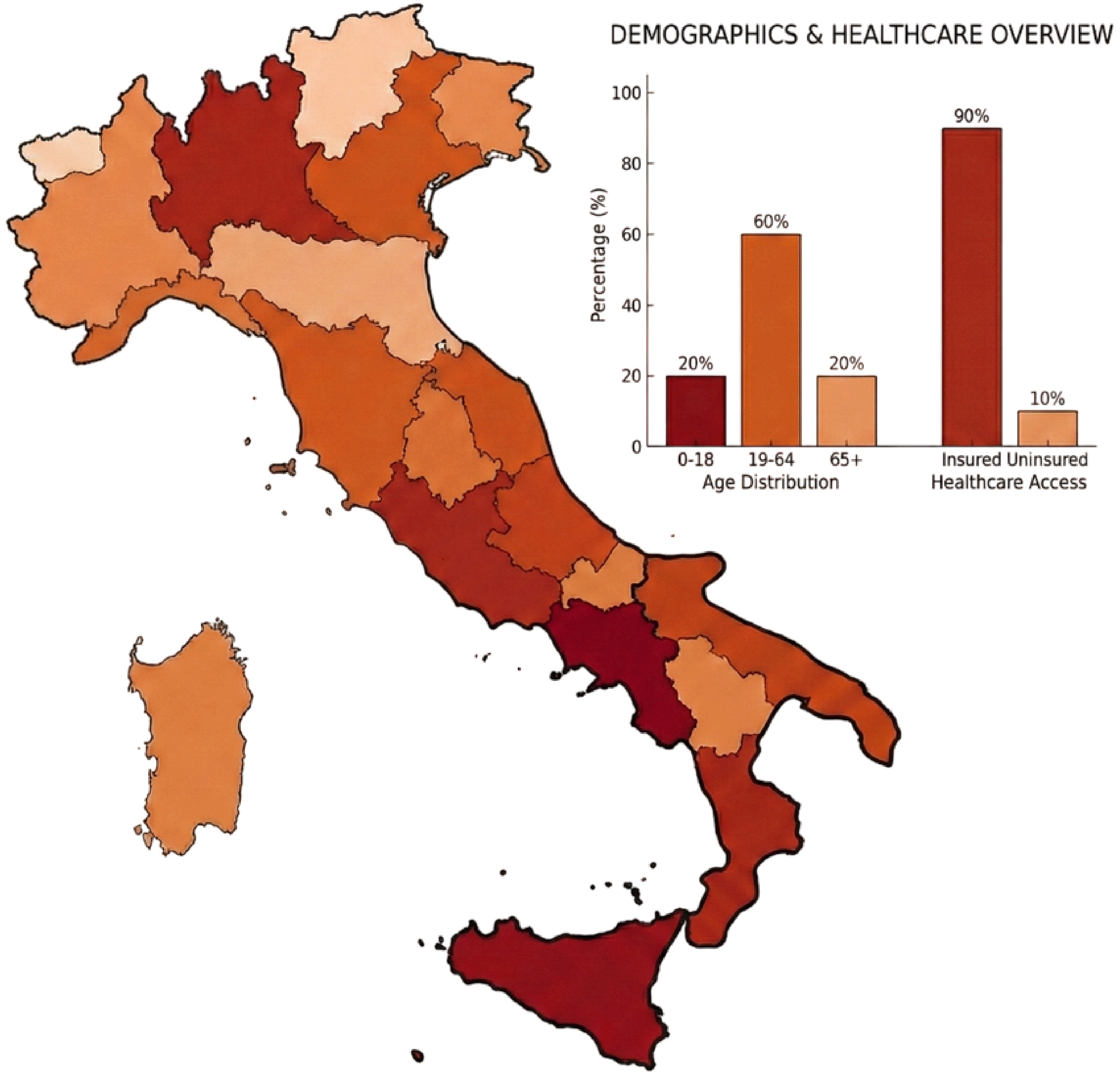
Sapienza Università di Roma

VA Project

Italian Regional Demographic and Healthcare

A. Pagnotta, F. Trionfetti, M. Sorrentini



Visual Analytics Course 2025/2026

Project Goal: A Unified Solution

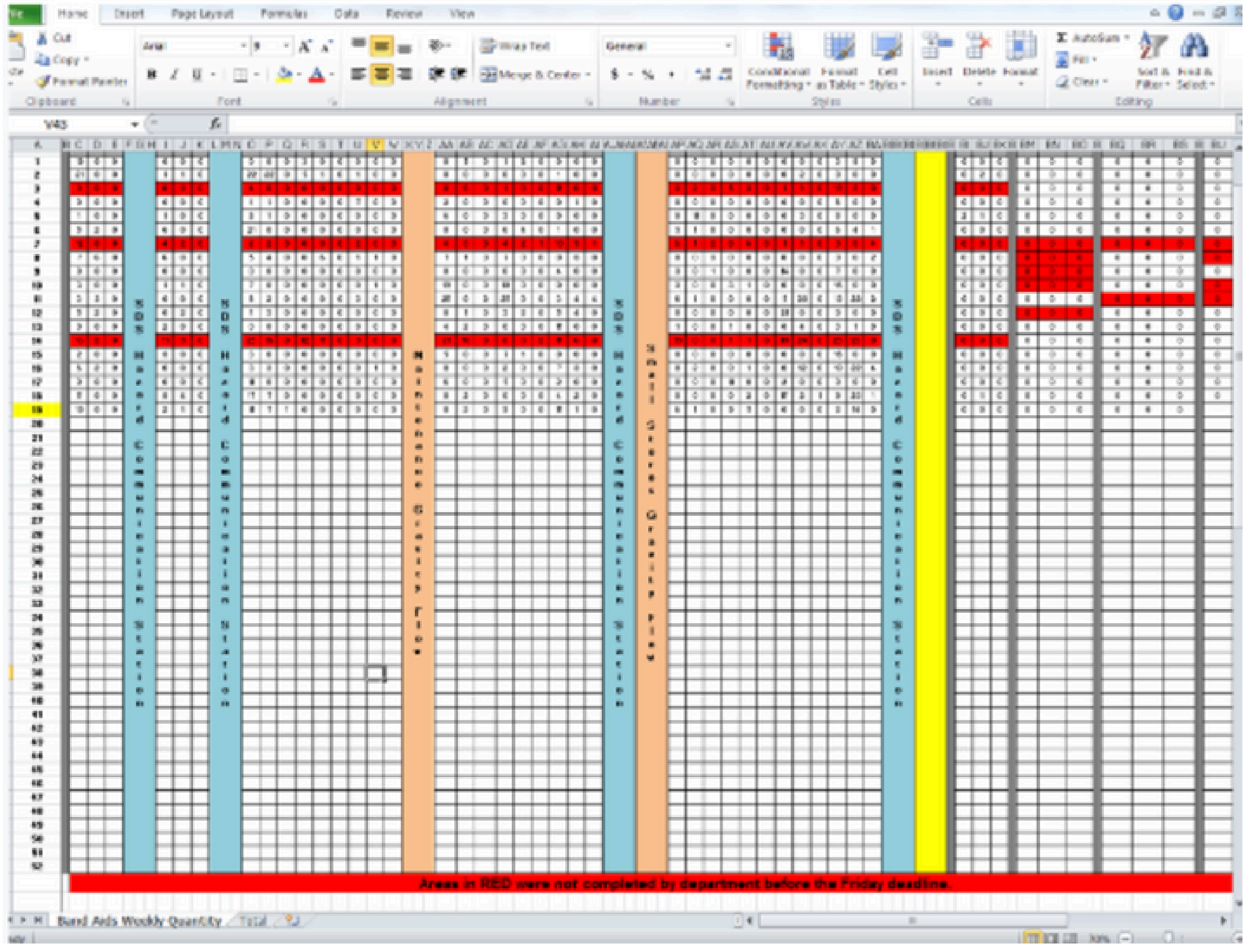
Build a Visual Analytics tool to enable a **Regional Analysis** and finding insights by visualizing the **demographic** and **healthcare** indicators across all Italian regions over a 50-year span using the data from ISTAT.

Key Approach: Integrate PCA, Clustering, and Visual Triggering to detect patterns without disrupting cognitive flow.

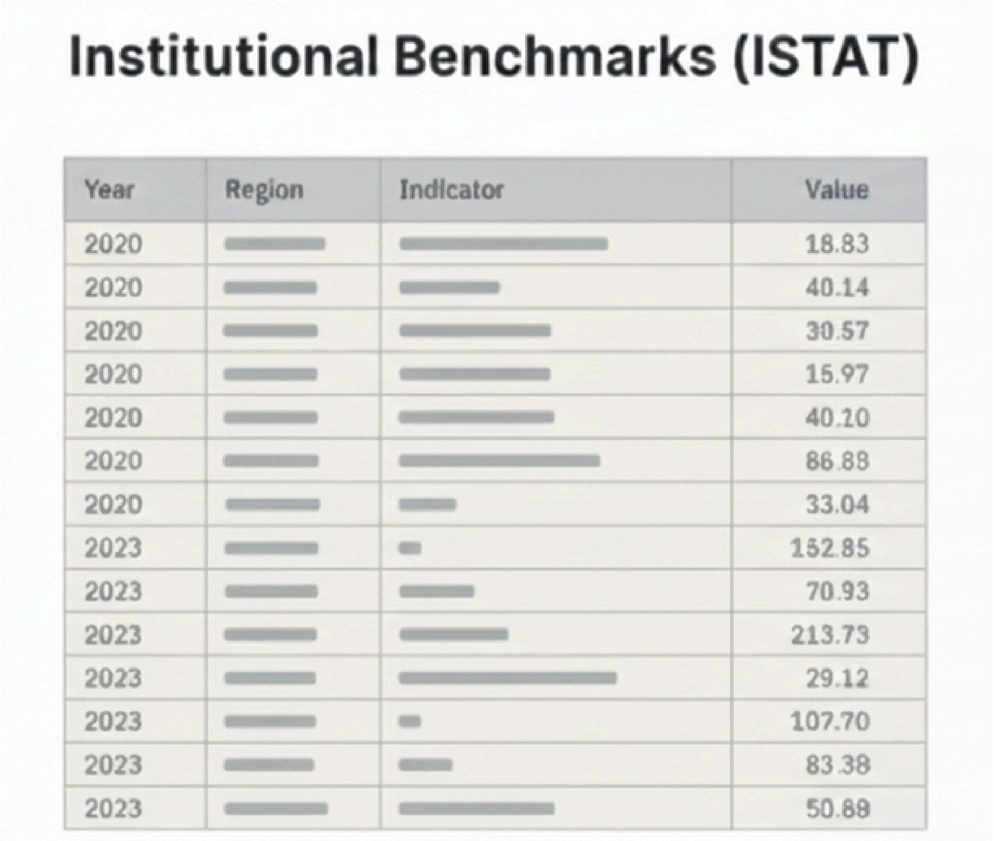
The Challenge: High Dimensionality

Demographic datasets offer rich information but they are difficult to explore due to complexity.

- **Univariate Focus**: Existing platforms force users to analyze one variable at a time.
- **Cognitive Load**: Analysts must manually toggle between views to find correlations.
- **Hidden Patterns**: Complex structural relationships remain invisible in spreadsheets.



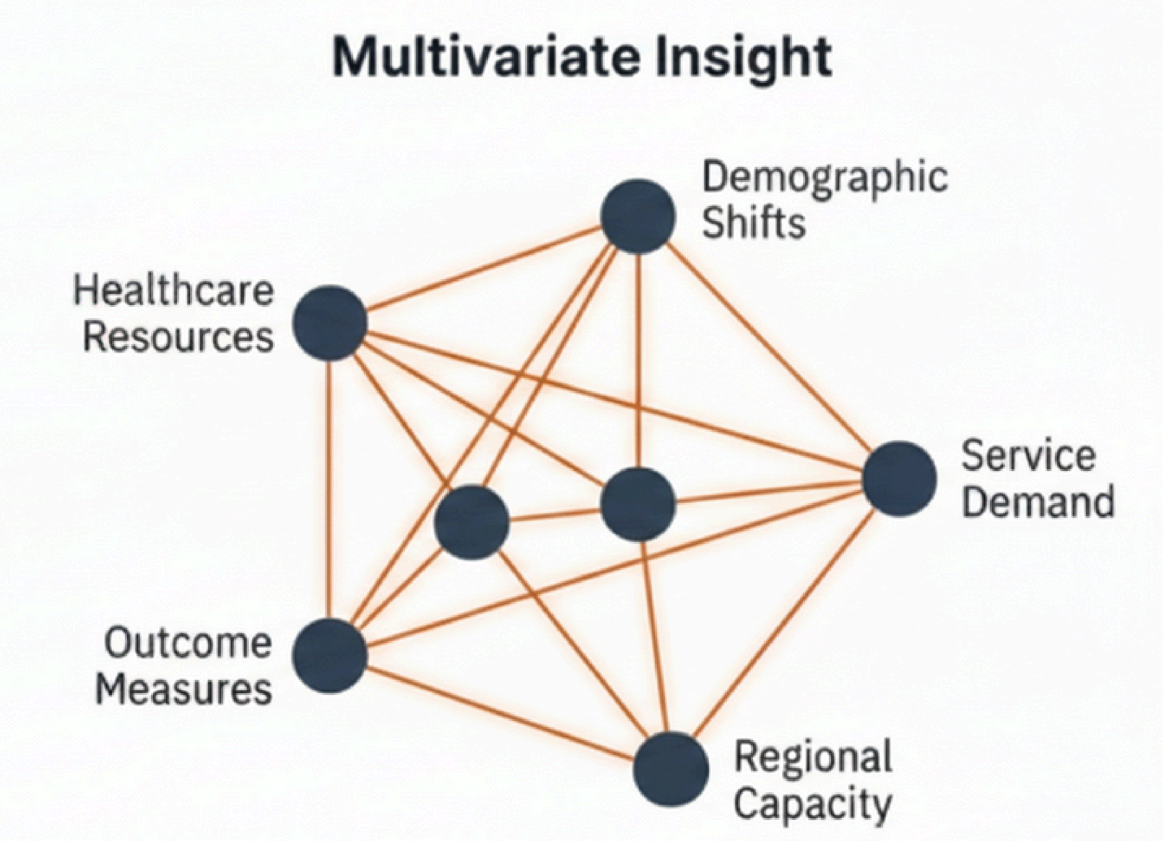
The Gap in Analytical Reasoning



Institutional Benchmarks (ISTAT)

Year	Region	Indicator	Value
2020			18.83
2020			40.14
2020			30.57
2020			15.97
2020			40.10
2020			86.88
2020			33.04
2023			152.85
2023			70.93
2023			213.73
2023			29.12
2023			107.70
2023			83.38
2023			50.88

Supports Retrieval: Users view one variable at a time (Univariate). Static and disconnected.



Supports Exploration: Users see the interplay between healthcare and demographic indicators.

Intended User & Analysis Goals

The **Target Audience** are **Demographic Analysts** and **Policy Makers** conducting preliminary investigations on high-dimensional datasets. This system lets them perform:

- **Multivariate Exploration:** Correlating heterogeneous indicators
- **Hypothesis Generation:** Identifying outliers and unexpected patterns to formulate and validate theories.
- **Spatiotemporal Integration:** Contextualizing phenomena both geographically and temporally.

More about the data

Sources

ISTAT: Demographics, historical records, and future projections.

Ministry of Health: Healthcare resources.

Dimensions

Spatiotemporal: 20 Regions (NUTS-2) x 48 Years (2002-2050).

Metrics: 32 quantitative indicators.

Total: ~32,000 observations.

Chained ARIMAX Forecasting

Future projections for healthcare indicators (up to 2050) are generated using a two-stage chained forecasting architecture.

Stage 1: ARIMA

Supply Side

Predicts structural resource evolution based on historical trends.

(e.g., Bed Density, Staff Density)

Stage 2: ARIMA

Human-Centric Outcomes

Uses Stage 1 outputs to forecast outcomes.

(e.g., Satisfaction Index)

Related Work & Innovation

Official Portals

ISTAT & **Ministry of Health** are reliable but **limited** to univariate analysis. They **lack** tools for cross-variable correlation.

Related Works

Drawing on the **state-of-the-art in Visual Analytics**, the dashboard design was developed by synthesizing the conceptual approaches of Jern et al., Diaz et al., and Dou et al.

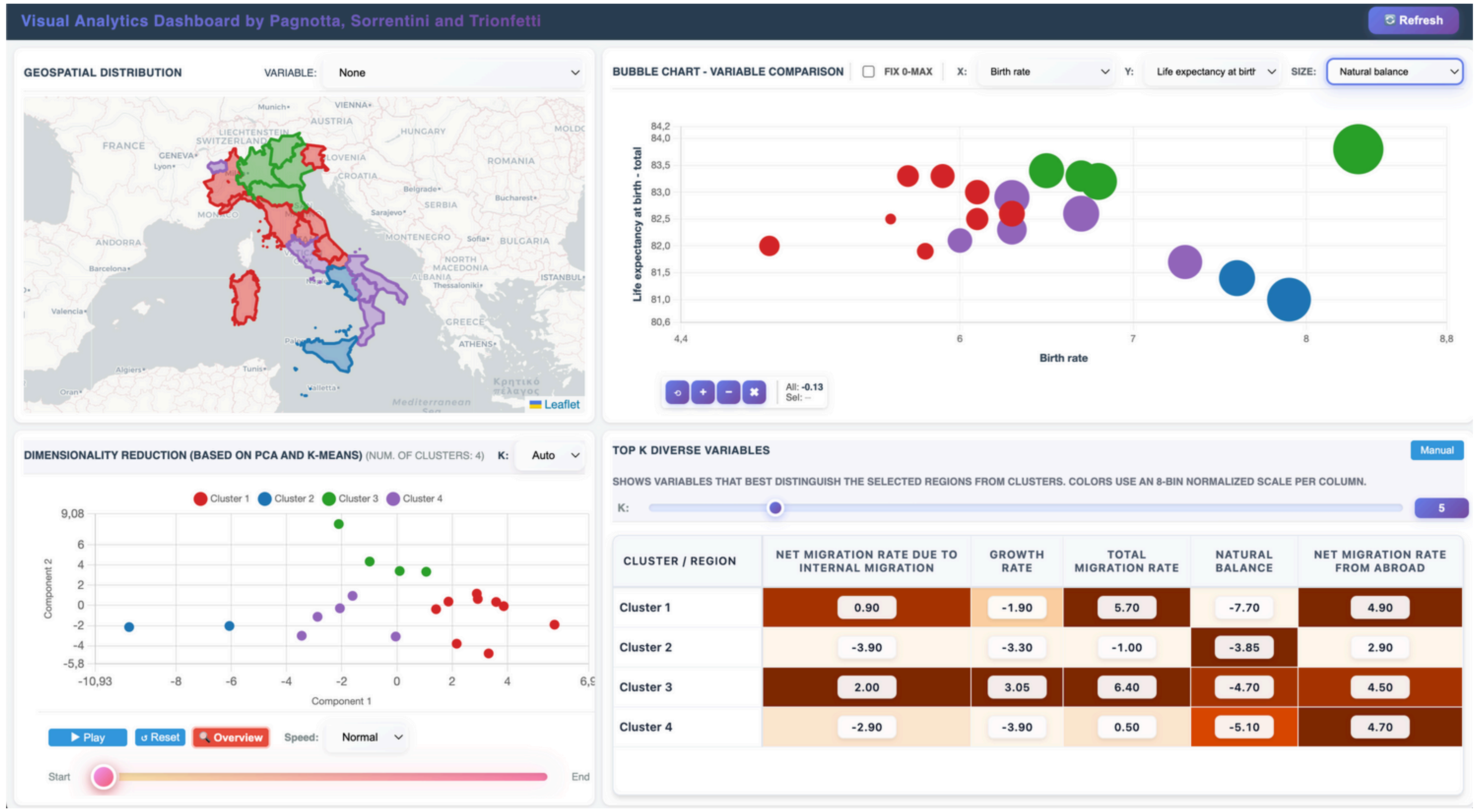
Gapminder

Inspired by Rosling's **Gapminder**, we use **Bubble Charts** with trails to visualize temporal evolution and cluster migration.

Design Overview

Detail (60% Width):

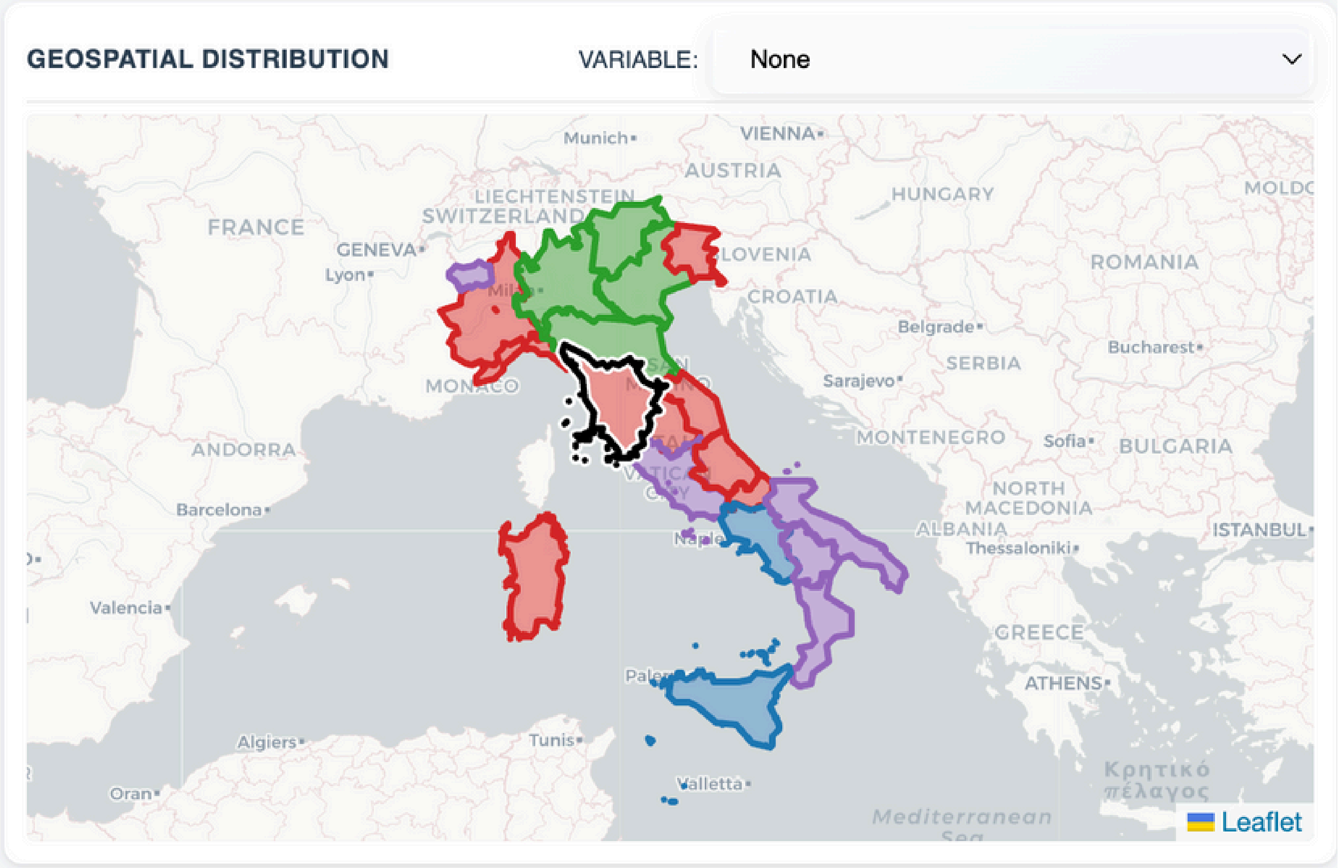
Bubble Chart and
Analytical Table



Context (40% Width):

Geospatial Map and
PCA Cluster Chart .

Geospatial Distribution



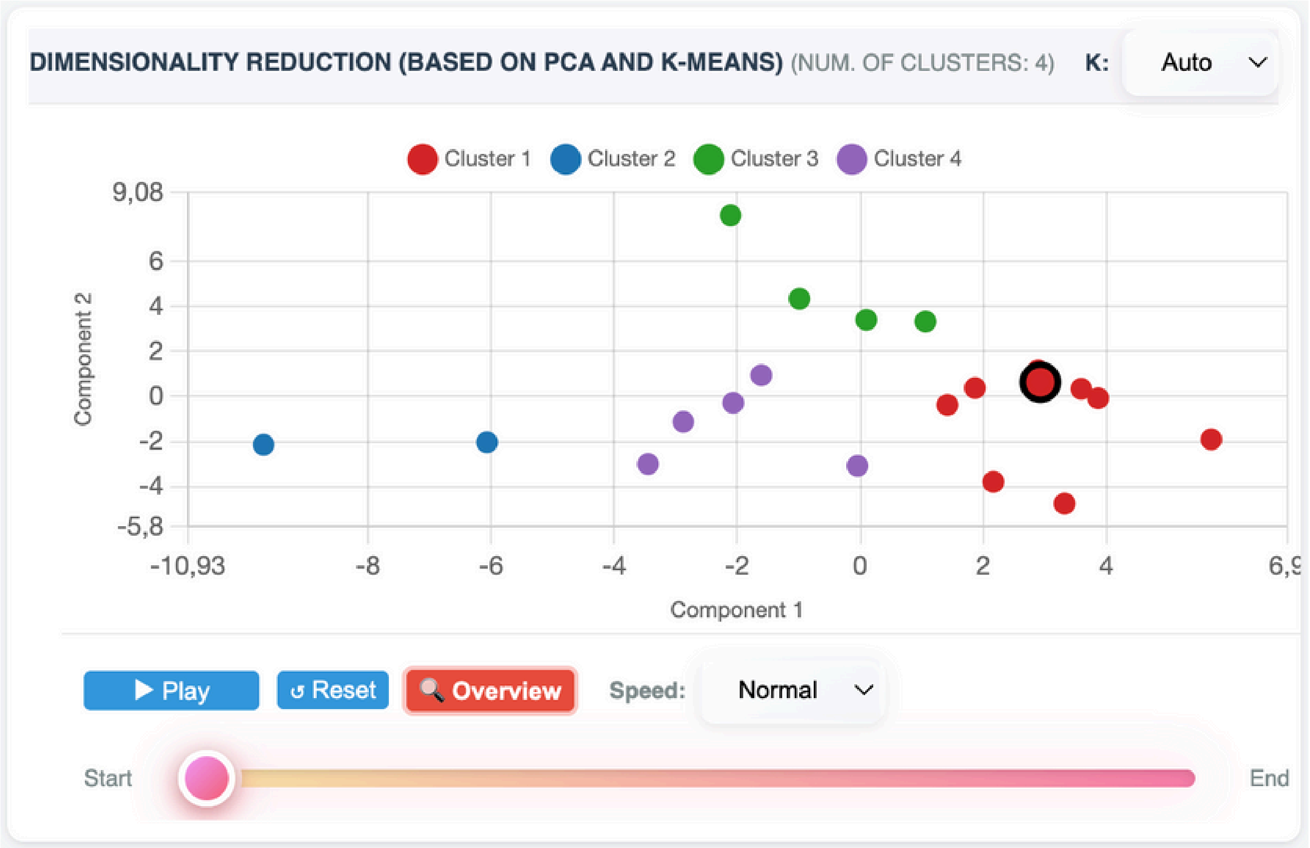
We choose to use a **Choropleth Map** (Leaflet.js). It serves as the spatial anchor for detecting territorial patterns. We have two working state:

- **Default State**: Region fill encodes Cluster ID (Qualitative scale).
- **Heatmap State**: Encodes specific indicator values (Sequential Orange scale).

Dimensionality Reduction (PCA)

The Dimensionality Reduction panel gives a Structural Overview of the 32 indicators into 2D space to reveal similarity patterns.

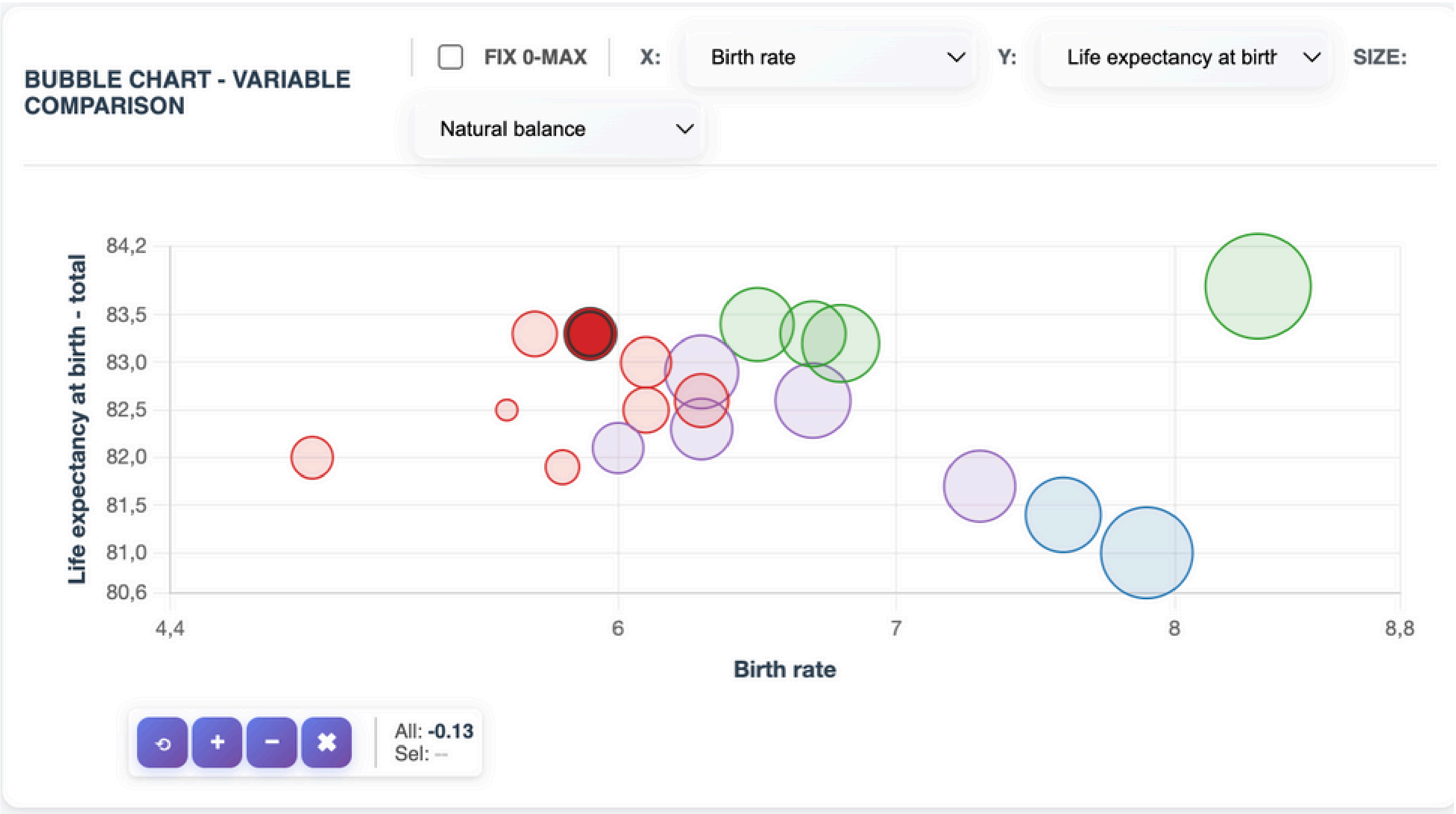
- **Logic**: Points closer together share similar demographic/healthcare profiles.
- **Clustering**: K-Means groups regions (Hue = Cluster ID).
- **Visual Triggering**: Users "brush" points to instantly identify what makes those regions unique (Analytical Table).



Variable Comparison

To visually compare different indicators we opted for a Gapminder-inspired Bubble Chart. This allowed to investigate direct relationships between specific indicators (some examples later in the insights section).

- **Encoding**: X/Y axes = Variables, Size = 3rd Variable, Color = Cluster.
- **Temporal Trails**: Selection triggers a historical trajectory (dashed line) to visualize evolution from 2002 to 2050.



Feature Differentiation

To further investigate the variables that best distinguish the **characteristics** of the regions, we implemented a table showing the the Top-K Ranking, to calculate a discriminativity score for all **32 variables** based on the selection of regions:

$$\text{Score} = \frac{|\mu_{\text{selected}} - \mu_{\text{rest}}|}{\sigma_{\text{global}}}$$

The table displays only the **Top-K** most diverse variables, providing instant semantic explanation for any visual cluster.

TOP K DIVERSE VARIABLES

Manual

SHOWS VARIABLES THAT BEST DISTINGUISH THE SELECTED REGIONS FROM CLUSTERS. COLORS USE AN 8-BIN NORMALIZED SCALE PER COLUMN.

K: 6

CLUSTER / REGION	BED DENSITY	LIFE EXPECTANCY AT BIRTH - MALES	TOTAL MIGRATION RATE	NET MIGRATION RATE FROM ABROAD	LIFE EXPECTANCY AT BIRTH - TOTAL	MEAN AGE AT CHILDBEARIN
Toscana	2.87	81.30	7.40	6.10	83.30	32.80
Cluster 1	3.38	80.40	5.40	4.85	82.55	32.50
Cluster 2	3.03	79.20	-1.00	2.90	81.20	31.70
Cluster 3	3.75	81.35	6.40	4.50	83.35	32.45

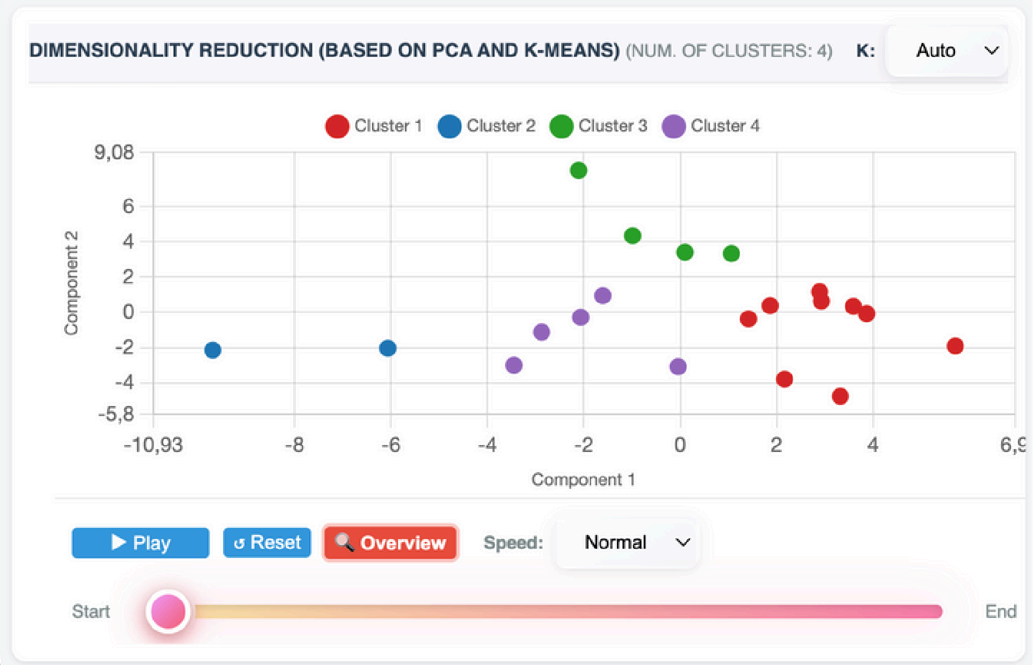
The Triggering Loop

We eliminate complex menus by utilizing direct manipulation as the primary driver of analysis:

1. Action (Brushing): Users draw a rectangular lasso around points of interest in the PCA plot.

2. Computation: The server executes a real-time variance pipeline (Z-score calculation) across 32 variables.

3. Result: The Analytical Table automatically reorders to display the most significant discriminants.



TOP K DIVERSE VARIABLES

Manual

SHOWS VARIABLES THAT BEST DISTINGUISH THE SELECTED REGIONS FROM CLUSTERS. COLORS USE AN 8-BIN NORMALIZED SCALE PER COLUMN.

K: 6

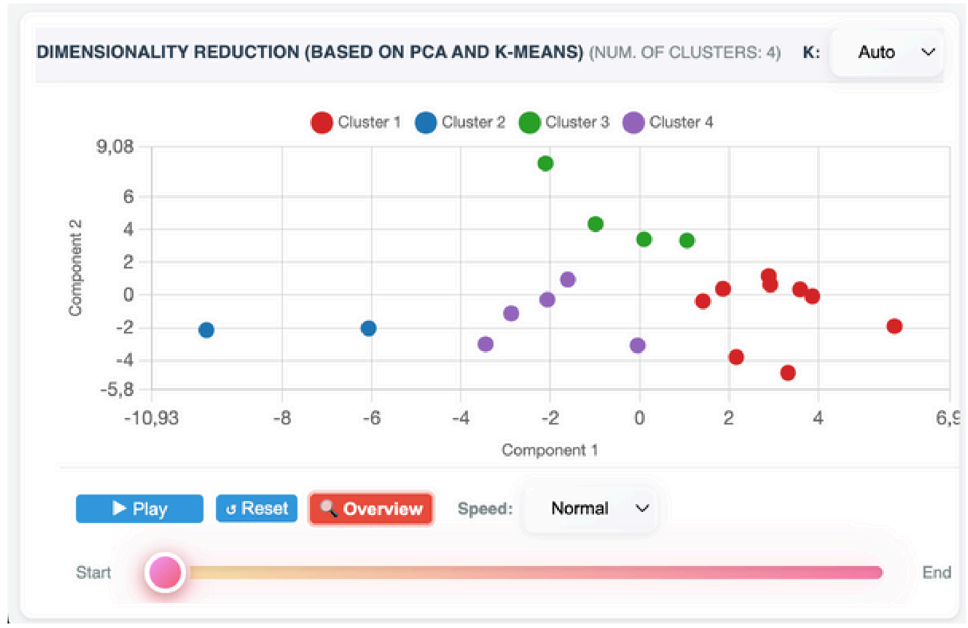
CLUSTER / REGION	BED DENSITY	LIFE EXPECTANCY AT BIRTH - MALES	TOTAL MIGRATION RATE	NET MIGRATION RATE FROM ABROAD	LIFE EXPECTANCY AT BIRTH - TOTAL	MEAN AGE AT CHILDBEARIN
Toscana	2.87	81.30	7.40	6.10	83.30	32.80
Cluster 1	3.38	80.40	5.40	4.85	82.55	32.50
Cluster 2	3.03	79.20	-1.00	2.90	81.20	31.70
Cluster 3	3.75	81.35	6.40	4.50	83.35	32.45

Context & Synchronization

Coordinated Highlighting (Linking): Visual continuity is guaranteed. Selecting a cluster in PCA instantly highlights the corresponding regions on the Map and their bubbles in the Chart.

Temporal Navigation: The system manages 2002-2050 data via an interactive player.

Linear Interpolation: Prevents "change blindness" by smoothing the animation frames, visualizing the velocity of demographic shifts.



Case Study 1: the North – South Divide

PCA reveals a distinct cluster for Southern regions.

Insight: Analytical Table identifies "Net Migration Rate" and "Growth Rate" as primary discriminants.

Forecast: The dashboard confirms no "catch-up" trend; growth rate disparities are projected to amplify by 2050.

CLUSTER / REGION	NET MIGRATION RATE DUE TO INTERNAL MIGRATION	TOTAL MIGRATION RATE	LIFE EXPECTANCY AT 65 - FEMALES	MARRIAGE RATE	LIFE EXPECTANCY AT BIRTH - FEMALES
Basilicata	-5.30	0.30	67.00	4.60	62.70
Calabria	-5.30	-0.30	67.90	5.30	62.80
Campania	-4.30	-1.40	67.20	4.40	63.80

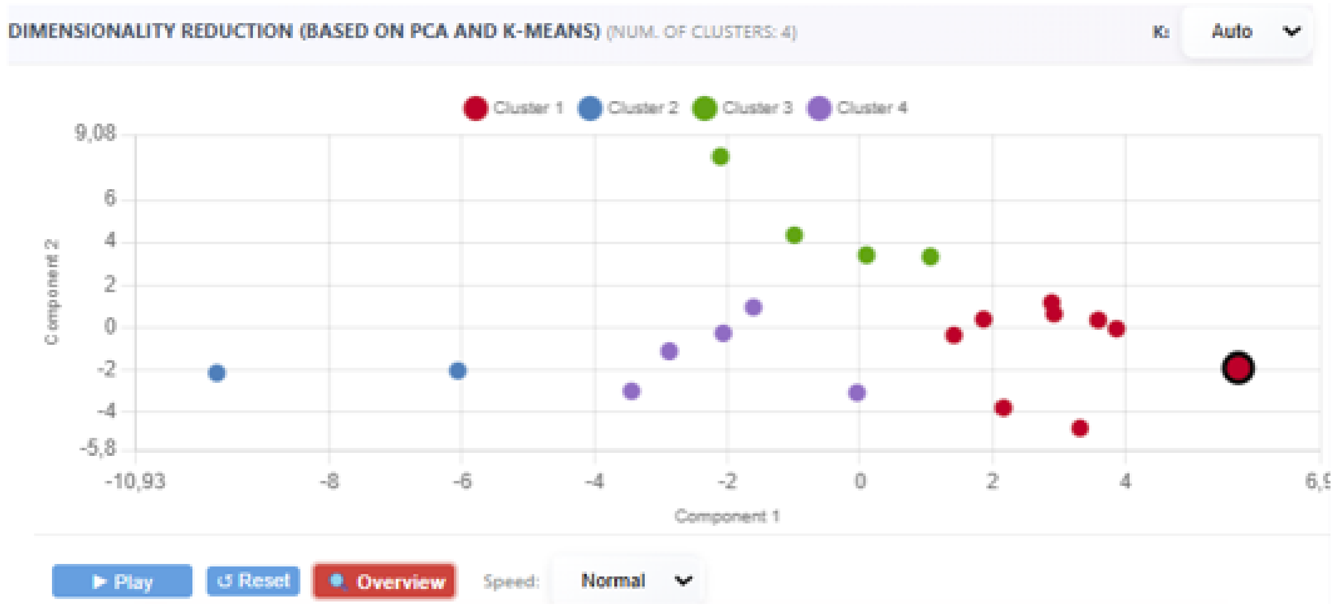
Case Study 2: the Liguria Anomaly

Historical Data: Liguria appears as a significant outlier due to extreme aging indices.

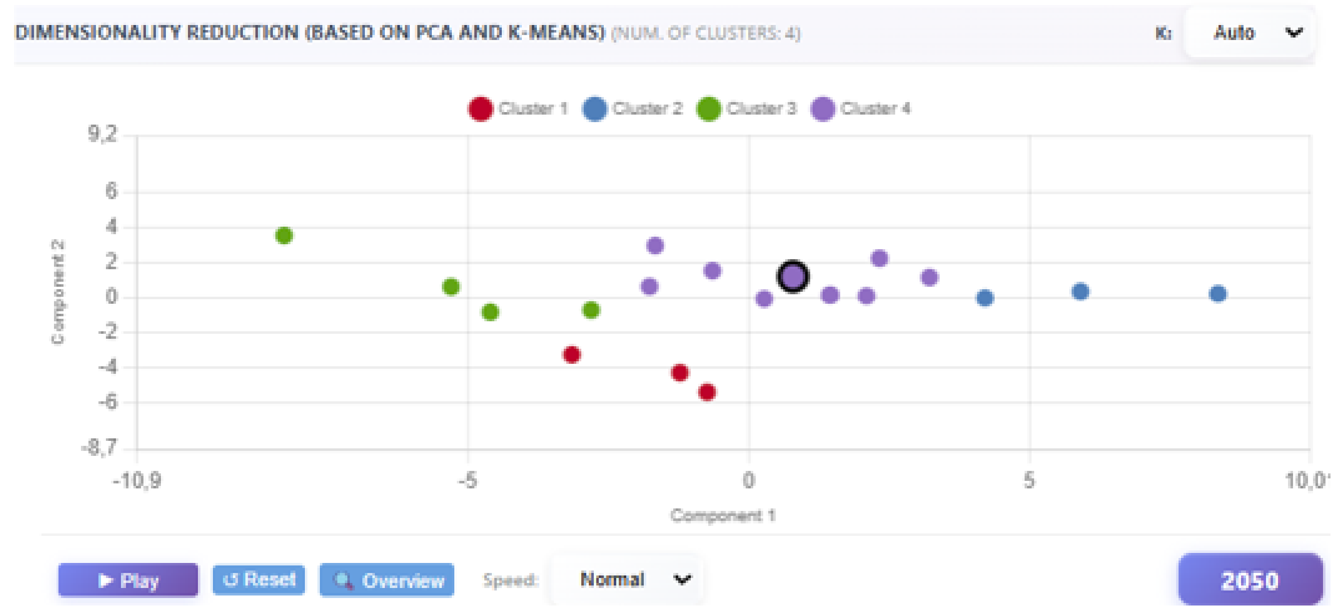
Projection (2024-2050): The region moves towards the main cluster.

Why? The "Feature Differentiation" engine reveals a shift: high immigration levels act as a structural rebalancer, mitigating historical isolation.

Overall scenario



2050 scenario

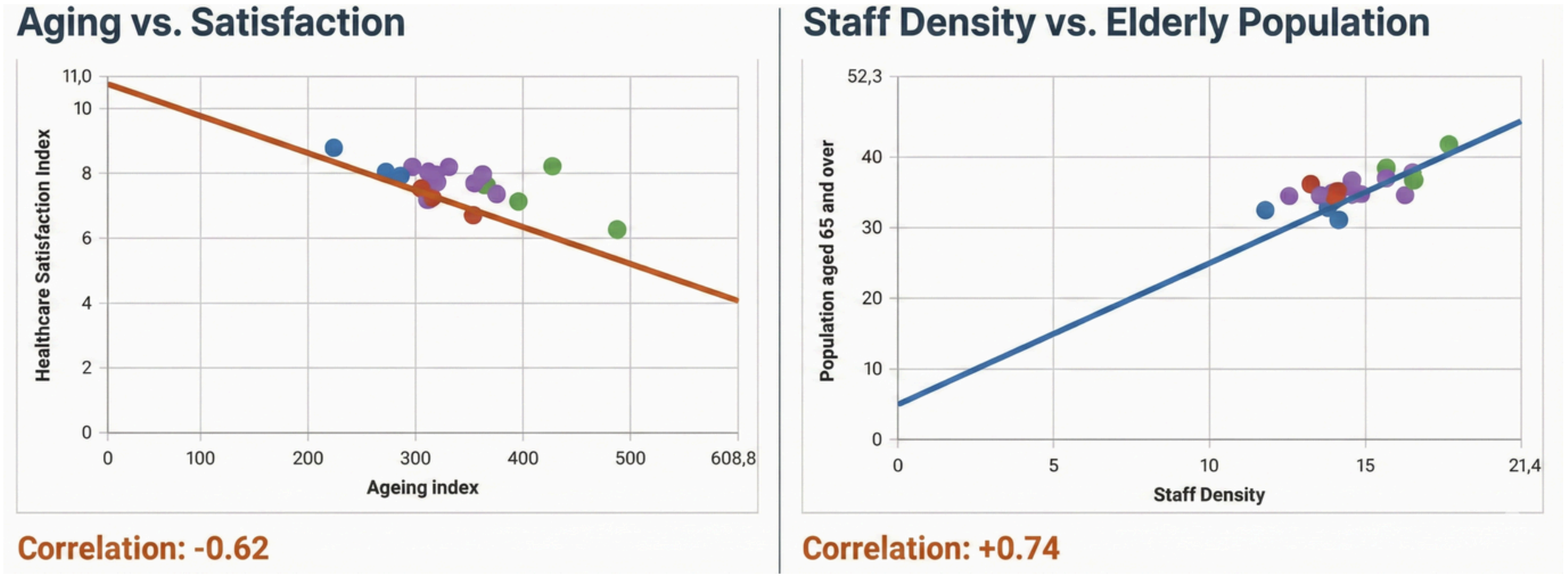


Case Study 3:
the Healtcare Sustanabilty Crisis

Analysis shows a negative correlation (-0.62) between Aging Index and Healthcare Satisfaction.

"System Crash": Simulations predict satisfaction drops below critical thresholds in regions like Sardegna by 2050.

Conclusion: Linear hiring is insufficient. An aggressive expansion of the geriatric workforce is mandatory.



Conclusion

Seamless transitions between spatial, structural, and longitudinal views without complex menus.

Successfully integrated 32 variables into a coherent dashboard for hypothesis generation.

Validated through case studies on regional disparities and healthcare forecasting.

Future Works

Granularity Expansion

Moving from NUTS-2 (Regions) to NUTS-3 (Provinces) to reveal local disparities currently flattened by aggregation.

&

WHAT IF...?

"What-If" Simulations

Integrating interactive simulation tools to allow policy makers to test the impact of specific interventions on future trends.

Thank you!

Project Repository: github.com/AntonioPagnotta/VA_project